TAREA #4

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Ejercicio 1

Inciso a)

Ecuación original: $y[n] - \frac{1}{2}y[n-1] = x[n]$

transformada z: $Y[z] - \frac{1}{2}Y[z]z^{-1} = X[z]$ → $H(z)$

↳ $Y[z]\left(1 - \frac{1}{2}z^{-1}\right) = X[z] \rightarrow \frac{Y[z]}{X[z]} = \frac{1}{1-\frac{1}{2}z^{-1}}$

Inciso b)

$z\{a^n u[n]\} = \frac{1}{1-az^{-1}} \rightarrow \frac{1}{1-1/2z^{-1}} = \left(\frac{1}{2}\right)^n u[n] = h[n]$

- Polo: Con este resultado me parece que el polo se encuentra en $\frac{1}{2}$, pues así se cumple que $1 - \frac{1}{2}z^{-1} = 0$.
- ROC: $|z| > 1/2$.

Inciso c)

$h[n] - 1/2\, h[n-1] = \delta[n] \rightarrow h[n] = \delta[n] + \frac{1}{2}h[n-1]$

- $h[0] = \frac{1}{2}(0) + 1 = 1$
- $h[1] = \frac{1}{2}(1) + 0 = 1/2$
- $h[2] = \frac{1}{2}\left(\frac{1}{2}\right) + 0 = 1/4$
- $h[3] = \frac{1}{2}\left(\frac{1}{4}\right) + 0 = 1/8$

← Por ser un sistema causal $h[n-1] = 0$

Si se comparan estos resultados con la formula analítica $h[n] = \left(\frac{1}{2}\right)^n u[n]$ es fácil notar que los valores serán los mismos.

Inciso d)

$z = e^{jw} \rightarrow H(z)\Big|_{z=e^{jw}} = \frac{1}{1 - \frac{1}{2}e^{-jw}}$

- Cuando $\omega = 0$ → $\frac{1}{1-\frac{1}{2}(1)} = \frac{1}{\frac{1}{2}} = 2$
- Cuando $\omega = \pi$ → $\frac{1}{1-\frac{1}{2}(-1)} = \frac{1}{\frac{3}{2}} = \frac{2}{3}$

Con estos resultados, la frecuencia 0 es la que amplifica más.

Ejercicio 2

Inciso a)

Ecuación original: $y[n] - \frac{1}{2}y[n-1] = x[n] + \frac{1}{2}x[n-1]$

transformada Z: $Y[z] - \frac{1}{2}Y[z]z^{-1} = X[z] + \frac{1}{2}X[z]z^{-1}$

$\hookrightarrow Y[z](1 - \frac{1}{2}z^{-1}) = X[z](1 + \frac{1}{2}z^{-1})$

$\hookrightarrow \frac{Y[z]}{X[z]} = H[z] = \frac{1 + \frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}}$

Inciso b)

$$H(z) = \frac{1 + \frac{1}{2}z^{-1} + \frac{1}{2}z^{-1} - \frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}} = \frac{1 + z^{-1} - \frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}}$$

$$= \frac{1 - \frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}} + \frac{z^{-1}}{1 - \frac{1}{2}z^{-1}} = 1 + \frac{z^{-1}}{1 - \frac{1}{2}z^{-1}}$$

Inciso c)

$$H(z) = 1 + (z^{-1})\left(\frac{1}{1 - \frac{1}{2}z^{-1}}\right) \rightarrow h[n] = \delta[n] + \left(\frac{1}{2}\right)^{n-1}u[n-1]$$

Inciso d)

$h[n] = \delta[n] + (\frac{1}{2})^{n-1}u[n-1]$

$\hookrightarrow$
- $h[0] = 1 + 0 = 1$
- $h[1] = 0 + (\frac{1}{2})^0 = 1$
- $h[2] = 0 + (\frac{1}{2})^1 = 1/2$
- $h[3] = 0 + (\frac{1}{2})^2 = 1/4$

$\therefore$ Ambos coinciden

$y[n] - \frac{1}{2}y[n-1] = x[n] + \frac{1}{2}x[n-1]$

$h[n] = \delta[n] + \frac{1}{2}\delta[n-1] + \frac{1}{2}h[n-1]$
- $h[0] = 1 + 0 + \frac{1}{2}(0) = 1$
- $h[1] = 0 + \frac{1}{2}(1) + \frac{1}{2}(1) = 1$
- $h[2] = 0 + 0 + \frac{1}{2}(1) = 1/2$
- $h[3] = 0 + 0 + \frac{1}{2}(\frac{1}{2}) = 1/4$

$$H(z)\Big|_{z=e^{j\omega}} = \frac{1 + \frac{1}{2}e^{-j\omega}}{1 - \frac{1}{2}e^{-j\omega}}$$

- Cuando $\omega_0 = 0$: $= \frac{1 + \frac{1}{2}(1)}{1 - \frac{1}{2}(1)} = \frac{\frac{3}{2}}{\frac{1}{2}} = 3$
- Cuando $\omega_0 = \pi$: $= \frac{1 + \frac{1}{2}(-1)}{1 - \frac{1}{2}(-1)} = \frac{1/2}{3/2} = \frac{1}{2}$

Ejercicio 3

Inciso a)

ecuación original: $y[n] - \frac{1}{4}y[n-1] - \frac{1}{8}y[n-2] = x[n]$

transformada z: $Y[z] - \frac{1}{4}Y[z]z^{-1} - \frac{1}{8}Y[z]z^{-2} = X[z]$

$\hookrightarrow Y[z]\left(1 - \frac{1}{4}z^{-1} - \frac{1}{8}z^{-2}\right) = X[z]$

$\hookrightarrow \frac{Y[z]}{X[z]} = \frac{1}{1 - \frac{1}{4}z^{-1} - \frac{1}{8}z^{-2}} = H(z)$

$$H[z] = \frac{1}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 + \frac{1}{4}z^{-1}\right)}$$

Inciso b)

$$H(z) = \frac{1}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 + \frac{1}{4}z^{-1}\right)} = \frac{A}{1 - \frac{1}{2}z^{-1}} + \frac{B}{1 + \frac{1}{4}z^{-1}}$$

$\hookrightarrow H(z) = 1 = A\left(1 + \frac{1}{4}z^{-1}\right) + B\left(1 - \frac{1}{2}z^{-1}\right) = A + \frac{1}{4}Az^{-1} + B - \frac{1}{2}Bz^{-1}$

$1 = (A+B) + z^{-1}\left(\frac{1}{4}A - \frac{1}{2}B\right)$

$\hookrightarrow A + B = 1 \longrightarrow A = 1 - B$

$\frac{1}{4}A - \frac{1}{2}B = 0 \longrightarrow \frac{1}{4}(1-B) - \frac{1}{2}B = 0 \rightarrow \frac{1}{4} - \frac{B}{4} - \frac{B}{2} = 0$

$\frac{1}{4} = \frac{2B + 4B}{8} \rightarrow \frac{1}{4} = \frac{3}{4}B$

$A + \frac{1}{3} = 1 \rightarrow A = 1 - \frac{1}{3} = \frac{2}{3}$ $\qquad \frac{1}{3} = B$

$$H(z) = \frac{\frac{2}{3}}{1 - \frac{1}{2}z^{-1}} + \frac{\frac{1}{3}}{1 + \frac{1}{4}z^{-1}}$$

Inciso c)

- $h[n] = \frac{2}{3}\left(\frac{1}{2}\right)^n u[n] + \frac{1}{3}\left(-\frac{1}{4}\right)^n u[n]$
- $h[n] = \delta[n] + \frac{1}{4}h[n-1] + \frac{1}{8}h[n-2]$

$\hookrightarrow h[0] = 1 + \frac{1}{4}(0) + \frac{1}{8}(0) = 1$

$\hookrightarrow h[1] = 0 + \frac{1}{4}(1) + \frac{1}{8}(0) = \frac{1}{4}$

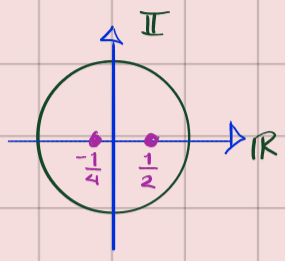
$\hookrightarrow h[2] = 0 + \frac{1}{4}\left(\frac{1}{4}\right) + \frac{1}{8}(1) = \frac{1}{16} + \frac{1}{8} = \frac{3}{16}$

$\hookrightarrow h[3] = 0 + \frac{1}{4}\left(\frac{3}{16}\right) + \frac{1}{8}\left(\frac{1}{4}\right) = \frac{3}{64} + \frac{1}{32} = \frac{5}{64}$

→ Inciso d)

$$H(z) = \frac{2/3}{1-\frac{1}{2}z^{-1}} + \frac{1/3}{1+\frac{1}{4}z^{-1}}$$ ← POLOS: $z = \frac{1}{2}$ y $z = -\frac{1}{4}$

Al ambos polos estar contenidos en el circulo unitario, el sistema es estable.



$$H(z)\Big|_{z=e^{j\omega}} = \frac{1}{1-\frac{1}{4}(e^{-j\omega}) - \frac{1}{8}(e^{-j\omega 2})}$$

• Cuando $\omega = 0$ → $= \frac{1}{1-\frac{1}{4}(1)-\frac{1}{8}(1)} = \frac{1}{1-3/8} = \frac{1}{5/8} = \frac{8}{5}$

Ejercicio 4

→ Inciso a)

ecuación original: $y[n] - \frac{1}{2}y[n-1] = x[n]$ ← $x[n] = \cos(\frac{\pi}{3}n)$

Transformada Z: $Y[z] - \frac{1}{2}Y[z]z^{-1} = X[z]$

↳ $Y[z](1-\frac{1}{2}z^{-1}) = X[z]$ → $\frac{Y[z]}{X[z]} = \frac{1}{1-\frac{1}{2}z^{-1}} = H(z)$

$$H(z)\Big|_{z=e^{j\omega}} = \frac{1}{1-\frac{1}{2}e^{-j\omega}}$$

→ Inciso b)

$e^{j\frac{\pi}{3}} = \cos\frac{\pi}{3} + j\,\text{sen}\,\frac{\pi}{3}$

$$H(e^{j\pi/3}) = \frac{1}{1-\frac{1}{2}(\cos\frac{\pi}{3} - j\,\text{sen}\,\frac{\pi}{3})} = \frac{1}{1-\frac{1}{2}(\frac{1}{2} - j\frac{\sqrt{3}}{2})} = \frac{1}{\frac{3}{4} + j\frac{\sqrt{3}}{4}}$$

↳ magnitud $= \frac{1}{\sqrt{(3/4)^2 + (\sqrt{3}/4)^2}} = \frac{1}{\sqrt{\frac{12}{16}}} = \frac{1}{\frac{\sqrt{3}}{2}} = \frac{2}{\sqrt{3}}$

↳ fase $= -\arctan\left(\frac{\sqrt{3}/4}{3/4}\right) = -\frac{\pi}{6}$ → Forma polar: $\frac{2}{\sqrt{3}}e^{-j\pi/6}$

Inciso c)

$x[n] = \cos(\omega_0 n) \rightarrow \frac{\pi}{3}$

$y_{ss}[n] = |H(e^{j\omega_0})| \cos(\omega_0 n + \phi)$

$y_{ss}[n] = \frac{2}{\sqrt{3}} \cos\left(\frac{\pi}{3} n - \frac{\pi}{6}\right)$

Inciso d)

- La frecuencia no cambia porque las exponenciales complejas son eigenfunciones de los sistemas LTI, ósea, solo alteran su amplitud y fase.
- La parte que se omitió pudo ser la respuesta transitoria, aunque me parece que dado mi $H(z)$, su polo sigue cayendo en el circulo unitario, lo que lo hace un sistema estable. resp. trans.

Ejercicio 5

Inciso a)

ecuación original: $y[n] - \frac{1}{4} y[n-2] = x[n] + x[n-1] - \frac{1}{4} x[n-2]$

transformada z: $Y[z] - \frac{1}{4} Y[z] z^{-2} = X[z] + X[z] z^{-1} - \frac{1}{4} X[z] z^{-2}$

$\hookrightarrow Y[z](1 - \frac{1}{4} z^{-2}) = X[z](1 + z^{-1} - \frac{1}{4} z^{-2})$

$\hookrightarrow \frac{Y[z]}{X[z]} = \frac{1 + z^{-1} - \frac{1}{4} z^{-2}}{1 - \frac{1}{4} z^{-2}}$

Inciso b)

$(1 + z^{-1} - \frac{1}{4} z^{-2}) - (1 - \frac{1}{4} z^{-2}) = z^{-1}$

$\hookrightarrow H(z) = 1 + \frac{z^{-1}}{1 - \frac{1}{4} z^{-2}}$

Inciso c)

$1 - \frac{1}{4} z^{-2} = (1 - \frac{1}{2} z^{-1})(1 + \frac{1}{2} z^{-1})$

$\hookrightarrow H(z) = \frac{z^{-1}}{(1 - \frac{1}{2} z^{-1})(1 + \frac{1}{2} z^{-1})} = \frac{A}{1 - \frac{1}{2} z^{-1}} + \frac{B}{1 + \frac{1}{2} z^{-1}}$

$z^{-1} = A(1 + \frac{1}{2} z^{-1}) + B(1 - \frac{1}{2} z^{-1}) = A + \frac{1}{2} A z^{-1} + B - \frac{1}{2} z^{-1} B$

$= (A + B) + z^{-1}(\frac{1}{2} A - \frac{1}{2} B)$

$A + B = 0 \rightarrow A = -B$

$\frac{1}{2} A - \frac{1}{2} B = 1$

$-\frac{B}{2}-\frac{B}{2}=1 \rightarrow -B=1 \rightarrow B=-1$, $A=1$

$$H(z)=1+\frac{1}{1-\frac{1}{2}z^{-1}}-\frac{1}{1+\frac{1}{2}z^{-1}}$$

* 1 aporta un impulso en $n=0$ porque su transformada inversa es $\delta[n]$, ósea, una respuesta instantánea en el origen.

→ Inciso d)

$$h[n]=\delta[n]+\left(\frac{1}{2}\right)^n u[n]-\left(-\frac{1}{2}\right)^n u[n]$$

↳ Por recurrencia: $h[n]=\frac{1}{4}h[n-2]+\delta[n]+\delta[n-1]-\frac{1}{4}\delta[n-2]$

- $h[0]=0+1+0-0=1$
- $h[1]=0+0+1-0=1$
- $h[2]=\frac{1}{4}(1)+0+0-\frac{1}{4}(1)=0$
- $h[3]=\frac{1}{4}(1)+0+0-0=\frac{1}{4}$
- $h[4]=\frac{1}{4}(0)+0+0-0=0$

→ Inciso e)

$$H(e^{j\omega})=\frac{1+e^{-j\omega}-\frac{1}{4}e^{-j2\omega}}{1-\frac{1}{4}e^{-j2\omega}}$$

- Cuando $\omega=0$: $=\frac{1+1-\frac{1}{4}}{1-\frac{1}{4}}=\frac{7/4}{3/4}=\frac{7}{3}$
- Cuando $\omega=\pi$: $=\frac{1-1-\frac{1}{4}(1)}{1-\frac{1}{4}(1)}=\frac{-1/4}{\frac{3}{4}}=-1/3$

Ejercicio 6

→ Inciso a)

ecuación original: $y[n]-\frac{1}{4}y[n-1]-\frac{1}{4}y[n-2]+\frac{1}{16}y[n-3]=x[n]$

transformada z: $Y[z]-\frac{1}{4}Y[z]z^{-1}-\frac{1}{4}Y[z]z^{-2}+\frac{1}{16}Y[z]z^{-3}=X[z]$

$$Y[z]\left(1-\frac{1}{4}z^{-1}-\frac{1}{4}z^{-2}+\frac{1}{16}z^{-3}\right)=X[z]$$

$$H[z]=\frac{Y[z]}{X[z]}=\frac{1}{1-\frac{1}{4}z^{-1}-\frac{1}{4}z^{-2}+\frac{1}{16}z^{-3}}$$

$$\left(1-\frac{1}{2}z^{-1}\right)\left(1+\frac{1}{2}z^{-1}\right)=1-\frac{1}{4}z^{-2} \rightarrow 1-\frac{1}{4}z^{-2}\left(1-\frac{1}{4}z^{-1}\right)=1-\frac{1}{4}z^{-1}-\frac{1}{4}z^{-2}+\frac{1}{16}z^{-3}$$

$$H(z)=\frac{1}{\left(1-\frac{1}{2}z^{-1}\right)\left(1+\frac{1}{2}z^{-1}\right)\left(1-\frac{1}{4}z^{-1}\right)}$$

Inciso b)

$$H(z) = \frac{A}{1-\frac{1}{2}z^{-1}} + \frac{B}{1+\frac{1}{2}z^{-1}} + \frac{C}{1-\frac{1}{4}z^{-1}}$$

$$H(z) \Rightarrow 1 = A\left(1+\tfrac{1}{2}z^{-1}\right)\left(1-\tfrac{1}{4}z^{-1}\right) + B\left(1-\tfrac{1}{2}z^{-1}\right)\left(1-\tfrac{1}{4}z^{-1}\right) + C\underbrace{\left(1-\tfrac{1}{2}z^{-1}\right)}_{(1+1/2\,z^{-1})}$$

$$1 = (A+B+C) + z^{-1}\left(\tfrac{1}{4}A - \tfrac{3}{4}B\right) + z^{-2}\left(-\tfrac{1}{8}A + \tfrac{1}{8}B - \tfrac{1}{4}C\right)$$

- $A+B+C = 1$
- $\frac{1}{4}A - \frac{3}{4}B = 0 \rightarrow A - 3B = 0 \rightarrow A = 3B$
- $-\frac{1}{8}A + \frac{1}{8}B - \frac{1}{4}C = 0 \rightarrow -\frac{1}{8}(3B) + \frac{1}{8}B - \frac{1}{4}C = 0$

  $-\frac{2}{8}B - \frac{1}{4}C = 0 \rightarrow C = -B$

$3B + B - B = 1 \rightarrow B = \frac{1}{3}$

$A = 3\left(\frac{1}{3}\right) = 1$

$C = -1/3$

$$H(z) = \frac{1}{1-\frac{1}{2}z^{-1}} + \frac{\frac{1}{3}}{1+\frac{1}{2}z^{-1}} - \frac{\frac{1}{3}}{1-\frac{1}{4}z^{-1}}$$

Inciso c)

$$h[n] = \left(\frac{1}{2}\right)^n u[n] + \frac{1}{3}\left(-\frac{1}{2}\right)^n u[n] - \frac{1}{3}\left(\frac{1}{4}\right)^n u[n]$$

↳ Por recurrencia: $h[n] = \frac{1}{4}h[n-1] + \frac{1}{4}h[n-2] - \frac{1}{16}h[n-3] + \delta[n]$

- $h[0] = 0 + 0 - 0 + 1 = 1$
- $h[1] = \frac{1}{4}(1) + 0 - 0 + 0 = 1/4$
- $h[2] = \frac{1}{4}\left(\frac{1}{4}\right) + \frac{1}{4}(1) - 0 + 0 = 5/16$
- $h[3] = \frac{1}{4}\left(\frac{5}{16}\right) + \frac{1}{4}\left(\frac{1}{4}\right) - \frac{1}{16}(1) + 0 = \frac{5}{64} + \frac{16}{64} - \frac{4}{64} = \frac{17}{64}$

- $h[0] = 1$
- $h[1] = \frac{1}{4}$
- $h[2] = \frac{5}{16}$ ✓
- $h[3] = \frac{17}{64}$

Inciso d)

- Dado la formula de $H(z)$ que consegui, los polos son: $z = \frac{1}{2}$, $z = -\frac{1}{2}$ y $z = 1/4$
- ROC causal: $|z| > \frac{1}{2}$
- Estabilidad: El sistema es estable, pues todos sus polos se encuentran en el circulo unitario.

Inciso e)

$$H(e^{j\omega}) = \frac{1}{(1-\frac{1}{2}e^{-j\omega})(1+\frac{1}{2}e^{-j\omega})(1-\frac{1}{4}e^{-j\omega})}$$

- Cuando $\omega = 0$ $= \frac{1}{(1-\frac{1}{2})(1+\frac{1}{2})(1-\frac{1}{4})} = \frac{1}{(\frac{1}{2})(\frac{3}{2})(\frac{3}{4})} = \frac{1}{\frac{9}{16}} = \frac{16}{9}$
- Cuando $\omega = \pi$ $= \frac{1}{(1-\frac{1}{2}(-1))(1+\frac{1}{2}(-1))(1-\frac{1}{4}(-1))} = \frac{1}{(\frac{3}{2})(\frac{1}{2})(\frac{5}{4})} = \frac{1}{\frac{15}{16}} = \frac{16}{15}$

Los polos con el decaimiento más lento son $z = \frac{1}{2}$ y $-\frac{1}{2}$, pues son los de mayor magnitud a diferencia de $z = 1/4$.